Intern Domain: Robotics & Automation

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Software: FANUC ROBOGUIDE

Controller target: FANUC R-30iB

Gun type: Servo spot-welding gun

Project video link: https://lnkd.in/p/guBHitkH



FANUC INDUSTRIAL ROBOT SPOT WELDING PROJECT

R-30iB Controller with Servo Gun – ROBOGUIDE Demonstration

1. Project Overview

This project is a two-point resistance spot-welding demonstration for a FANUC R-30iB robot equipped with a servo welding gun. The robot approaches defined weld locations on a clamped sheet-metal workpiece. At each taught point, the configured servo-gun SPOT instruction performs the programmed gun motion and weld sequence. The intended environment is a virtual FANUC ROBOGUIDE workcell.

2. Objectives

Create a simulated spot-welding cell; understand robot approach and weld-point teaching; use the FANUC servo-gun SPOT instruction; configure tool and user frames; review gun setup and welding schedules; and validate reach, clearance, sequence and simulated I/O before considering any physical application.

3. Workcell Design

The proposed cell contains one FANUC R-30iB robot, one servo spot-welding gun, two sheet-metal components, a locating fixture, a base/table and suitable safety guarding. Two weld locations are used to demonstrate a short sequence. The gun must be configured as a servo-gun axis in the controller. The robot tool frame, payload, gun TCP, close direction and user frame must match the selected virtual cell.

4. Servo-Gun Programming Concept

The FANUC servo-gun manual describes a SPOT instruction format using start distance (SD), pressure schedule (P), part thickness (t), weld schedule (S) and end distance (ED). A representative instruction is SPOT[SD=1,P=1,t=2.0,S=1,ED=1]. The values in this example are placeholders only. They identify schedule slots and an illustrative thickness; they are not welding recommendations. Configure schedules from a qualified, validated welding procedure.

5. Program Sequence

The supplied LS template moves to a safe start position, approaches weld point 1, executes a servo-gun SPOT instruction, retracts, moves to weld point 2, executes the second SPOT instruction, retracts and returns to the safe position. P[1] and P[3] are approach points; P[2] and P[4] are weld points; P[5] is a safe start/retract point. All positions in the LS file are example coordinates and must be retaught.

6. ROBOGUIDE Implementation Steps

1. Create a workcell with a controller revision compatible with the installed R-30iB Servo Gun and spot-welding options.
2. Add the robot, servo gun, sheet-metal workpiece and fixture. Check the gun model and axis group.
3. Verify servo-gun configuration, mastering, gun close direction, tool frame, payload and TCP.

4. Configure pressure, start-distance, end-distance and weld schedules using approved process data.
5. Teach the safe start, approach and weld positions in the actual virtual workcell.
6. Load or recreate the supplied LS program. Verify the SPOT instruction parameters and schedule references.
7. Use permitted simulation, tryout or weld-I/O simulation modes. Check gun closure, sequence completion and opening.
8. Check reach, collisions, electrode alignment, fixture clearance and alarms. Save the workcell and capture evidence.

7. Testing and Expected Outcome

The simulation should show the robot moving safely between two weld locations and initiating the configured servo-gun sequence at each location. Review the robot path, gun movement, schedule references, weld-complete feedback and alarms. Record only the results actually observed in ROBOGUIDE. Add screenshots of the cell, program, gun setup and completed simulation to the final submission.

8. Safety and Engineering Limitations

This is an educational starter example, not a production-ready or safety-certified program. Exact instruction syntax and screens depend on controller software revision and installed options. Weld current, weld time, squeeze/hold time, electrode force, pressure, tip geometry and material parameters must be established by a qualified welding/process engineer using a validated welding procedure. Do not run this template on physical equipment without approved commissioning, guarding, interlocks, risk assessment, emergency-stop checks, calibration and authorized supervision.

9. Conclusion

This project introduces the main elements of FANUC servo-gun spot-welding automation: workcell design, servo-gun configuration, taught robot positions, schedule-based SPOT instructions and simulation-based validation. Completing the workcell and recording the actual simulation will provide practical evidence of robotics and automation skills.

10. Submission Evidence Checklist

☐ ROBOGUIDE workcell screenshot showing robot, gun and fixture.

☐ Servo Gun setup screen and configured schedule references.

☐ Program listing showing SPOT instructions.

☐ Screenshots or video of both simulated weld cycles.

☐ Saved ROBOGUIDE workcell and final corrected LS program.

References

FANUC Robot Series R-30iB Servo Gun Operator's Manual, B-83264EN (consult the revision matching the installed controller).

FANUC Robot Series R-30iB Controller Spot Welding Function Operator's Manual, B-83284EN-4 (consult the revision matching the installed controller).

Program Model:

```
/PROG  KUGANRAJ_SPOT _WELDING
/ATTR
OWNER            = MNEDITOR;
COMMENT                  = "25.5.26";
PROG_SIZE        = 13088;
CREATE           = DATE 26-05-25  TIME 09:49:20;
MODIFIED         = DATE 26-05-25  TIME 09:49:20;
FILE_NAME        = RBC_35JP;
VERSION                  = 0;
LINE_COUNT       = 246;
MEMORY_SIZE    = 13380;
PROTECT                  = READ_WRITE;
TCD:  STACK_SIZE         = 0,
    TASK_PRIORITY        = 50,
    TIME_SLICE    = 0,
    BUSY_LAMP_OFF        = 0,
    ABORT_REQUEST        = 0,
    PAUSE_REQUEST        = 0;
DEFAULT_GROUP            = 1,1,*,*,*;
CONTROL_CODE = 00000000 00000000;
/APPL
 SPOT : TRUE ;
/MN
   1:  !FMA R402 ;
```

```
 2:  ;
 3:  UFRAME_NUM=0 ;
 4:  UTOOL_NUM=1 ;
 5:  PAYLOAD[1] ;
 6:  ;
 7:  OVERRIDE=R[49:Override] ;
 8:  ;
 9:J PR[2:HOME POS NEW] 50% FINE    ;
10:  ;
11:  DO[36:NIZ TO R404-1]=ON ;
12:  DO[44:NIZ TO R404-2]=ON ;
13:  ;
14:  DO[1:CYCLE COMPLETE]=OFF ;
15:  DO[2:WELD COMP 1 TO S040]=OFF ;
16:  DO[98:ISO CONTACTER]=OFF ;
17:  ;
18:  DO[20:NIZ TO S040]=ON ;
19:  ;
20:  ;
21:J P[1] 75% CNT70    ;
22:J P[2] 75% CNT70    ;
23:J P[3] 75% CNT70    ;
24:J P[4] 75% CNT70    ;
25:J P[5] 75% CNT50 BREAK    ;
26:  ;
27:  WAIT DI[1:READY OK 1FROM S040]=ON    ;
28:  WAIT DI[20:NIZ FROM S040]=ON    ;
29:  DO[20:NIZ TO S040]=OFF ;
```

```
30:  ;
31:  DO[98:ISO CONTACTER]=ON ;
32:  WAIT DI[97:ISO CONTACTOR]=ON    ;
33:  ;
34:  TIMER[1]=RESET ;
35:  TIMER[1]=START ;
36:  //WAIT DI[44:NIZ FROM R404-2]=ON    ;
37:  //DO[44:NIZ TO R404-2]=OFF ;
38:  //WAIT DI[44:NIZ FROM R404-2]=ON    ;

39:  ;
40:  ;
41:  ;
42:  ;
43:  ;
44:L P[6] 1500mm/sec CNT70    ;
45:L P[7] 1500mm/sec CNT50    ;
46:L P[8] 1500mm/sec CNT50    ;
47:L P[9] 1500mm/sec CNT70    ;
48:L P[10] 2000mm/sec CNT80
  :  SPOT[SD=3,P=12,S=6,ED=3]    ;
49:L P[11] 2000mm/sec CNT80
  :  SPOT[SD=3,P=12,S=6,ED=3]    ;
50:L P[12] 1500mm/sec CNT70    ;
51:L P[13] 1500mm/sec CNT60    ;
52:L P[146] 2000mm/sec CNT70    ;
53:L P[147] 2000mm/sec CNT70    ;
54:L P[161] 2000mm/sec CNT80
```

: SPOT[SD=3,P=12,S=6,ED=3] ;

55:L P[82] 2000mm/sec CNT80

: SPOT[SD=3,P=12,S=6,ED=3] ;

56:L P[162] 1500mm/sec CNT70 ;

57:L P[146] 1500mm/sec CNT60 ;

58:L P[13] 1500mm/sec CNT60 ;

59:L P[14] 2000mm/sec CNT60 ;

60:L P[15] 2000mm/sec CNT60 ;

61:L P[16] 200mm/sec CNT70 ;

62:L P[17] 1500mm/sec CNT70 ;

63:L P[98] 1500mm/sec CNT60 ;

64: ;

65:L P[18] 1500mm/sec CNT50 ;

66:L P[19] 1500mm/sec CNT70 ;

67: ;

68: DO[44:NIZ TO R404-2]=OFF ;

69: ;

70: ;

71:L P[20] 1500mm/sec CNT30 ;

72:L P[21] 2000mm/sec CNT30 ;

73:L P[22] 2000mm/sec CNT80

: SPOT[SD=2,P=12,S=6,ED=2] ;

74:L P[54] 2000mm/sec CNT80

: SPOT[SD=2,P=12,S=6,ED=2] ;

75:L P[132] 2000mm/sec CNT60 ;

76:L P[23] 2000mm/sec CNT80

: SPOT[SD=2,P=12,S=6,ED=2] ;

77:L P[133] 500mm/sec CNT60 ;

78:L P[24] 2000mm/sec CNT80

: SPOT[SD=3,P=12,S=6,ED=3] ;

79:L P[25] 2000mm/sec CNT80

: SPOT[SD=3,P=12,S=6,ED=3] ;

80:L P[26] 2000mm/sec CNT40 ;

81:L P[27] 2000mm/sec CNT40 ;

82:L P[151] 2000mm/sec CNT80

: SPOT[SD=3,P=12,S=6,ED=3] ;

83:L P[158] 500mm/sec CNT60 ;

84:L P[93] 2000mm/sec CNT80

: SPOT[SD=3,P=11,S=5,ED=3] ;

85: ;

86:L P[28] 2000mm/sec CNT80

: SPOT[SD=3,P=12,S=6,ED=3] ;

87:L P[29] 2000mm/sec CNT80

: SPOT[SD=3,P=12,S=6,ED=3] ;

88:L P[78] 2000mm/sec CNT80

: SPOT[SD=3,P=12,S=6,ED=3] ;

89:L P[79] 2000mm/sec CNT60 ;

90:L P[80] 2000mm/sec CNT80

: SPOT[SD=3,P=12,S=6,ED=3] ;

91:L P[87] 2000mm/sec CNT60 ;

92: ;

93:L P[92] 2000mm/sec CNT40 ;

94: ;

95: ;

96:L P[30] 500mm/sec CNT60 ;

97:L P[32] 2000mm/sec CNT80

```
  :  SPOT[SD=3,P=10,S=0,ED=3]    ;
 98:L P[33] 2000mm/sec CNT80
  :  SPOT[SD=3,P=10,S=6,ED=3]    ;
 99:L P[34] 2000mm/sec CNT80
  :  SPOT[SD=3,P=10,S=6,ED=3]    ;
100:L P[37] 2000mm/sec CNT80
  :  SPOT[SD=3,P=10,S=6,ED=3]    ;
101:L P[84] 2000mm/sec CNT80
  :  SPOT[SD=3,P=10,S=6,ED=3]    ;
102:L P[86] 2000mm/sec CNT80
  :  SPOT[SD=3,P=10,S=6,ED=3]    ;
103:   ;
104:L P[49] 2000mm/sec CNT80
  :  SPOT[SD=3,P=10,S=6,ED=3]    ;
105:L P[55] 1500mm/sec CNT60    ;
106:L P[104] 2000mm/sec CNT40    ;
107:L P[106] 2000mm/sec CNT20    ;
108:L P[39] 2000mm/sec CNT80
  :  SPOT[SD=3,P=15,S=7,ED=3]    ;
109:L P[105] 1500mm/sec CNT20    ;
110:L P[40] 2000mm/sec CNT5    ;
111:L P[42] 2000mm/sec CNT80
  :  SPOT[SD=3,P=15,S=7,ED=3]    ;

112:L P[107] 2000mm/sec CNT5    ;
113:L P[43] 2000mm/sec CNT40    ;
114:  DO[36:NIZ TO R404-1]=OFF ;
115:   ;
```

116:L P[44] 2000mm/sec CNT50 ;

117:L P[41] 2000mm/sec CNT5 ;

118:L P[45] 2000mm/sec CNT80

: SPOT[SD=3,P=15,S=7,ED=3] ;

119:L P[108] 2000mm/sec CNT60 ;

120:L P[46] 2000mm/sec CNT50 ;

121:L P[47] 2000mm/sec CNT60 ;

122:L P[109] 2000mm/sec CNT5 ;

123:L P[48] 2000mm/sec CNT80

: SPOT[SD=3,P=15,S=7,ED=3] ;

124:L P[110] 2000mm/sec CNT5 ;

125:L P[126] 500mm/sec FINE ;

126:L P[85] 2000mm/sec CNT80

: SPOT[SD=3,P=15,S=7,ED=3] ;

127:L P[156] 2000mm/sec CNT80

: SPOT[SD=4,P=10,S=12,ED=4] ;

128:L P[35] 1500mm/sec CNT60 ;

129:L P[36] 2000mm/sec CNT80

: SPOT[SD=3,P=13,S=6,ED=3] ;

130:L P[38] 1500mm/sec CNT60 ;

131: ;

132: ;

133:L P[50] 1500mm/sec CNT30 BREAK ;

134: ;

135:L P[51] 1500mm/sec CNT60 ;

136: WAIT DI[44:NIZ FROM R404-2]=ON ;

137: DO[44:NIZ TO R404-2]=OFF ;

138: WAIT DI[44:NIZ FROM R404-2]=ON ;

```
139:  ;
140:L P[52] 2000mm/sec CNT60    ;
141:L P[56] 2000mm/sec CNT40    ;
142:L P[57] 2000mm/sec CNT40    ;
143:L P[73] 1500mm/sec CNT60    ;
144:L P[74] 2000mm/sec CNT80
  :  SPOT[SD=3,P=15,S=7,ED=3]    ;
145:  ;
146:L P[58] 2000mm/sec CNT80
  :  SPOT[SD=3,P=15,S=7,ED=3]    ;
147:  ;
148:L P[59] 2000mm/sec CNT50    ;
149:L P[60] 2000mm/sec CNT50    ;
150:L P[61] 2000mm/sec CNT50    ;
151:L P[76] 1500mm/sec CNT60    ;
152:L P[77] 2000mm/sec CNT80
  :  SPOT[SD=3,P=14,S=9,ED=3]    ;
153:  ;
154:L P[152] 2000mm/sec CNT80
  :  SPOT[SD=3,P=14,S=2,ED=3]    ;
155:L P[81] 2000mm/sec CNT80
  :  SPOT[SD=3,P=14,S=9,ED=3]    ;
156:L P[62] 2000mm/sec CNT80
  :  SPOT[SD=3,P=15,S=5,ED=3]    ;
157:L P[53] 2000mm/sec CNT80
  :  SPOT[SD=3,P=15,S=5,ED=3]    ;
158:L P[83] 2000mm/sec CNT80
  :  SPOT[SD=3,P=15,S=5,ED=3]    ;
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```
159:  ;
160:L P[63] 2000mm/sec CNT80
   :  SPOT[SD=3,P=15,S=5,ED=3]    ;
161:L P[75] 2000mm/sec CNT80
   :  SPOT[SD=3,P=15,S=5,ED=3]    ;
162:  ;
163:L P[64] 2000mm/sec CNT80
   :  SPOT[SD=3,P=15,S=7,ED=3]    ;
164:L P[111] 2000mm/sec CNT40    ;
165:L P[65] 2000mm/sec CNT60    ;
166:L P[112] 2000mm/sec CNT50    ;
167:L P[113] 2000mm/sec CNT60    ;
168:L P[114] 2000mm/sec CNT40    ;
169:L P[66] 2000mm/sec CNT50    ;
170:L P[153] 2000mm/sec CNT80
   :  SPOT[SD=3,P=15,S=7,ED=3]    ;
171:L P[67] 2000mm/sec CNT80
   :  SPOT[SD=3,P=15,S=7,ED=3]    ;
172:L P[115] 1500mm/sec CNT40    ;
173:L P[68] 2000mm/sec CNT80
   :  SPOT[SD=3,P=11,S=5,ED=3]    ;
174:L P[116] 1500mm/sec CNT60    ;
175:  ;
176:L P[69] 2000mm/sec CNT80
   :  SPOT[SD=3,P=11,S=5,ED=3]    ;
177:L P[117] 1500mm/sec CNT60    ;
178:L P[70] 2000mm/sec CNT80
   :  SPOT[SD=3,P=11,S=5,ED=3]    ;
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```
179:L P[118] 1500mm/sec CNT60    ;
180:L P[71] 2000mm/sec CNT80
  : SPOT[SD=3,P=11,S=5,ED=3]    ;
181:  ;
182:  ;
183:L P[119] 1500mm/sec CNT40    ;
184:  DO[44:NIZ TO R404-2]=OFF ;
185:L P[72] 1900mm/sec CNT40    ;
186:L P[122] 1500mm/sec CNT60    ;
187:L P[123] 1500mm/sec CNT60    ;
188:L P[124] 1500mm/sec FINE    ;
189:L P[125] 1500mm/sec FINE    ;
190:  DO[1:CYCLE COMPLETE]=PULSE,1.0sec ;
191:  DO[2:WELD COMP 1 TO S040]=PULSE,1.0sec ;
192:  ;
193:L P[127] 1500mm/sec FINE    ;
194:L P[128] 1500mm/sec CNT60    ;
195:L P[129] 1500mm/sec CNT60    ;
196:L P[131] 1500mm/sec CNT60    ;
197:  ;
198:  ;
199:  ;
200:  DO[1:CYCLE COMPLETE]=PULSE,1.0sec ;
201:  DO[2:WELD COMP 1 TO S040]=PULSE,1.0sec ;
202:  ;
203:  DO[36:NIZ TO R404-1]=ON ;
204:  ;
205:  ;
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```
206:  //DO[1:CYCLE COMPLETE]=PULSE,1.0sec ;
207:  //DO[2:WELD COMP 1 TO S040]=PULSE,1.0sec ;
208:  ;
209:  DO[44:NIZ TO R404-2]=ON ;
210:  DO[44:NIZ TO R404-2]=ON ;
211:  ;
212:  ;
213:  DO[20:NIZ TO S040]=ON ;
214:  ;
215:  DO[98:ISO CONTACTER]=OFF ;
216:  ;
217:  TIMER[1]=STOP ;
218:  ;
219:J P[99] 75% CNT70    ;
220:J P[100] 75% CNT70    ;
221:J P[101] 75% CNT30    ;
222:  ;
223:  ;
224:  ;
225:  DO[33:NIZ TO R401-1]=ON ;
226:  DO[35:NIZ TO R403-1]=ON ;
227:  DO[36:NIZ TO R404-1]=ON ;
228:  DO[41:NIZ TO R401-2]=ON ;
229:  DO[43:NIZ TO R403-2]=ON ;
230:  DO[44:NIZ TO R404-2]=ON ;
231:  ;
232:  DO[1:CYCLE COMPLETE]=OFF ;
233:  DO[2:WELD COMP 1 TO S040]=OFF ;
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```
 234:  DO[3:WELD COMP 2 TO S040]=OFF ;
 235:  DO[4:WELD COMP 3 TO S040]=OFF ;
 236:  DO[98:ISO CONTACTER]=OFF ;
 237:  ;
 238:  DO[20:NIZ TO S040]=ON ;
 239:  ;
 240:  CALL _TEST_CT    ;
 241:  ;
 242:J PR[2:HOME POS NEW] 50% FINE    ;
 243:  ;
 244:  UFRAME_NUM=0 ;
 245:  UTOOL_NUM=1 ;
 246:  PAYLOAD[1] ;
/POS
P[1]{
  GP1:
        UF : 0, UT : 1,
        J1=  -107.017 deg,          J2=    33.914 deg,          J3=   -15.628 deg,
        J4=     1.007 deg, J5=   -72.769 deg,          J6=   108.934 deg
  GP2:
        UF : 0, UT : 1,
        J1=    -3.000  mm
};
P[2]{
  GP1:
        UF : 0, UT : 1,
        J1=  -108.003 deg,          J2=    28.024 deg,          J3=    -9.389 deg,
        J4=      .952 deg,  J5=   -78.990 deg,          J6=   110.036 deg
```

GP2:

UF : 0, UT : 1,

J1= -3.000 mm

};

P[3]{

GP1:

UF : 0, UT : 1,

J1= -77.810 deg, J2= -10.412 deg, J3= 5.978 deg,

J4= -.909 deg, J5= -101.588 deg, J6= 33.104 deg

GP2:

UF : 0, UT : 1,

J1= -3.000 mm

};

P[4]{

GP1:

UF : 0, UT : 1,

J1= -70.695 deg, J2= -18.691 deg, J3= 6.276 deg,

J4= -28.809 deg, J5= -101.238 deg, J6= -22.000 deg

GP2:

UF : 0, UT : 1,

J1= -3.000 mm

};

P[5]{

GP1:

UF : 0, UT : 1,

J1= -64.634 deg, J2= -35.133 deg, J3= 13.309 deg,

J4= -43.055 deg, J5= -112.963 deg, J6= -38.931 deg

GP2:

```
        UF : 0, UT : 1,
        J1=  -240.000  mm
};
P[6]{
  GP1:
        UF : 0, UT : 1,
        J1=   -57.543 deg,        J2=   -43.321 deg,        J3=   -11.303 deg,
        J4=   -62.422 deg,        J5=  -116.295 deg,        J6=   -26.970 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -240.000  mm
};
P[7]{
  GP1:
        UF : 0, UT : 1,
        J1=   -59.960 deg,        J2=   -21.607 deg,        J3=   -39.628 deg,
        J4=   -73.464 deg,        J5=  -108.956 deg,        J6=   -26.594 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -240.000  mm
};
P[8]{
  GP1:
        UF : 0, UT : 1,
        J1=   -57.252 deg,        J2=   -10.976 deg,        J3=   -45.585 deg,
        J4=   -73.710 deg,        J5=  -105.363 deg,        J6=   -21.205 deg
  GP2:
        UF : 0, UT : 1,
```

J1= -240.000 mm

};

P[9]{

GP1:

UF : 0, UT : 1,

J1= -55.984 deg, J2= 2.400 deg, J3= -57.741 deg,

J4= -72.032 deg, J5= -101.015 deg, J6= -9.514 deg

GP2:

UF : 0, UT : 1,

J1= -240.000 mm

};

P[10]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1406.636 mm, Y = -373.020 mm, Z = -339.169 mm,

W = 89.995 deg, P = 69.000 deg, R = -35.005 deg

GP2:

UF : 0, UT : 1,

J1= -34.000 mm

};

P[11]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1368.724 mm, Y = -348.676 mm, Z = -346.932 mm,

W = 89.990 deg, P = 69.000 deg, R = -35.014 deg

GP2:

UF : 0, UT : 1,

J1= -34.569 mm

};

P[12]{

GP1:

UF : 0, UT : 1,

J1= -55.984 deg, J2= 2.400 deg, J3= -57.741 deg,

J4= -72.032 deg, J5= -101.015 deg, J6= -9.514 deg

GP2:

UF : 0, UT : 1,

J1= -240.000 mm

};

P[13]{

GP1:

UF : 0, UT : 1,

J1= -58.700 deg, J2= -18.516 deg, J3= -46.675 deg,

J4= -85.741 deg, J5= -100.761 deg, J6= -21.950 deg

GP2:

UF : 0, UT : 1,

J1= -240.000 mm

};

P[14]{

GP1:

UF : 0, UT : 1,

J1= -59.960 deg, J2= -21.607 deg, J3= -39.628 deg,

J4= -73.464 deg, J5= -108.956 deg, J6= -26.594 deg

GP2:

UF : 0, UT : 1,

J1= -240.000 mm

};

P[15]{

GP1:

UF : 0, UT : 1,

J1= -57.543 deg, J2= -43.321 deg, J3= -11.303 deg,

J4= -62.422 deg, J5= -116.295 deg, J6= -26.970 deg

GP2:

UF : 0, UT : 1,

J1= -240.000 mm

};

P[16]{

GP1:

UF : 0, UT : 1,

J1= 16.160 deg, J2= -44.623 deg, J3= 10.487 deg,

J4= -3.823 deg,J5= -103.039 deg, J6= -69.940 deg

GP2:

UF : 0, UT : 1,

J1= -240.000 mm

};

P[17]{

GP1:

UF : 0, UT : 1,

J1= 25.567 deg, J2= -46.376 deg, J3= -19.081 deg,

J4= 4.809 deg, J5= -67.542 deg, J6= -74.613 deg

GP2:

UF : 0, UT : 1,

J1= -240.000 mm

};

P[18]{

```
  GP1:
        UF : 0, UT : 1,
        J1=    18.598 deg,          J2=   -20.016 deg,          J3=   -31.898 deg,
        J4=     9.408 deg, J5=   -63.478 deg,          J6=   -68.563 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -240.000  mm
};
P[19]{
  GP1:
        UF : 0, UT : 1,
        J1=    10.912 deg,          J2=   -30.424 deg,          J3=   -33.087 deg,
        J4=    17.608 deg,          J5=   -54.657 deg,          J6=   -45.476 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -240.000  mm
};
P[20]{
  GP1:
        UF : 0, UT : 1,
        J1=    10.563 deg,          J2=    -7.991 deg, J3=   -42.242 deg,
        J4=    30.033 deg,          J5=   -55.879 deg,          J6=   -55.301 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -240.000  mm
};
P[21]{
  GP1:
```

```
        UF : 0, UT : 1,
        J1=   10.888 deg,        J2=  -14.607 deg,        J3=  -34.456 deg,
        J4=   27.024 deg,        J5=  -62.047 deg,        J6=  -50.864 deg
  GP2:
        UF : 0, UT : 1,
        J1= -240.000  mm
};
P[22]{
  GP1:
        UF : 0, UT : 1,          CONFIG : 'N U T, 0, 0, 0',
        X =  1655.285  mm,       Y =  -391.334  mm,       Z =  -335.140  mm,
        W =   -20.249 deg,       P =   -12.942 deg,       R =   -30.194 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -16.206  mm
};
P[23]{
  GP1:
        UF : 0, UT : 1,          CONFIG : 'N U T, 0, 0, 0',
        X =  1740.695  mm,       Y =  -369.503  mm,       Z =  -344.536  mm,
        W =   -24.566 deg,       P =    -8.762 deg, R =   -18.355 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -44.000  mm
};
P[24]{
  GP1:
        UF : 0, UT : 1,          CONFIG : 'N U T, 0, 0, 0',
```

X = 1786.217 mm, Y = -366.851 mm, Z = -342.900 mm,

W = -27.605 deg, P = -9.608 deg, R = -17.686 deg

GP2:

UF : 0, UT : 1,

J1= -51.920 mm

};

P[25]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1825.328 mm, Y = -368.697 mm, Z = -344.565 mm,

W = -25.803 deg, P = -8.875 deg, R = -17.962 deg

GP2:

UF : 0, UT : 1,

J1= -54.395 mm

};

P[26]{

GP1:

UF : 0, UT : 1,

J1= 9.766 deg, J2= -1.828 deg, J3= -36.932 deg,

J4= 25.060 deg, J5= -60.310 deg, J6= -54.731 deg

GP2:

UF : 0, UT : 1,

J1= -89.262 mm

};

P[27]{

GP1:

UF : 0, UT : 1,

J1= 11.338 deg, J2= 4.707 deg, J3= -37.186 deg,

J4= 25.531 deg, J5= -66.271 deg, J6= -52.150 deg

GP2:

UF : 0, UT : 1,

J1= -137.515 mm

};

P[28]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1866.534 mm, Y = -330.616 mm, Z = -358.697 mm,

W = -24.656 deg, P = -5.536 deg, R = -23.494 deg

GP2:

UF : 0, UT : 1,

J1= -55.646 mm

};

P[29]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1866.441 mm, Y = -303.117 mm, Z = -368.524 mm,

W = -22.927 deg, P = -4.452 deg, R = -23.664 deg

GP2:

UF : 0, UT : 1,

J1= -93.724 mm

};

P[30]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1661.299 mm, Y = 149.667 mm, Z = -463.721 mm,

W = .393 deg, P = 2.833 deg, R = .795 deg

GP2:

UF : 0, UT : 1,

J1= -163.540 mm

};

P[32]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1661.826 mm, Y = 149.601 mm, Z = -398.174 mm,

W = .393 deg, P = 2.833 deg, R = .795 deg

GP2:

UF : 0, UT : 1,

J1= -13.558 mm

};

P[33]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1658.049 mm, Y = 113.481 mm, Z = -399.356 mm,

W = .094 deg, P = 3.387 deg, R = -1.848 deg

GP2:

UF : 0, UT : 1,

J1= -43.984 mm

};

P[34]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1655.881 mm, Y = 60.585 mm, Z = -398.995 mm,

W = -4.460 deg, P = 2.958 deg, R = -7.494 deg

GP2:

UF : 0, UT : 1,

J1= -57.695 mm

};

P[35]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1389.496 mm, Y = -195.675 mm, Z = -435.526 mm,

W = -7.056 deg, P = -3.252 deg, R = -33.393 deg

GP2:

UF : 0, UT : 1,

J1= -86.347 mm

};

P[36]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1380.376 mm, Y = -194.338 mm, Z = -397.168 mm,

W = -7.065 deg, P = -3.220 deg, R = -33.406 deg

GP2:

UF : 0, UT : 1,

J1= -51.915 mm

};

P[37]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1659.393 mm, Y = 5.645 mm, Z = -396.986 mm,

W = -4.683 deg, P = .415 deg, R = -20.882 deg

GP2:

UF : 0, UT : 1,

J1= -21.228 mm

};

P[38]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1375.055 mm, Y = -197.211 mm, Z = -436.975 mm,

W = -7.922 deg, P = -2.365 deg, R = -35.157 deg

GP2:

UF : 0, UT : 1,

J1= -80.143 mm

};

P[39]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1660.549 mm, Y = -249.780 mm, Z = -387.868 mm,

W = -11.362 deg, P = -1.797 deg, R = -15.084 deg

GP2:

UF : 0, UT : 1,

J1= -62.621 mm

};

P[40]{

GP1:

UF : 0, UT : 1,

J1= 5.770 deg, J2= -11.721 deg, J3= -31.998 deg,

J4= 15.535 deg, J5= -64.814 deg, J6= -37.000 deg

GP2:

UF : 0, UT : 1,

J1= -89.291 mm

};

P[41]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1554.848 mm, Y = -324.703 mm, Z = -419.715 mm,

W = -16.046 deg, P = -6.756 deg, R = -22.037 deg

GP2:

UF : 0, UT : 1,

J1= -140.000 mm

};

P[42]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1660.318 mm, Y = -312.487 mm, Z = -372.362 mm,

W = -15.121 deg, P = -1.805 deg, R = -17.915 deg

GP2:

UF : 0, UT : 1,

J1= -16.691 mm

};

P[43]{

GP1:

UF : 0, UT : 1,

J1= 7.479 deg, J2= -14.463 deg, J3= -34.838 deg,

J4= 21.213 deg, J5= -60.516 deg, J6= -39.360 deg

GP2:

UF : 0, UT : 1,

J1= -140.000 mm

};

P[44]{

GP1:

UF : 0, UT : 1,

J1= 9.666 deg, J2= -19.765 deg, J3= -34.708 deg,

J4= 21.425 deg, J5= -60.812 deg, J6= -44.537 deg

GP2:

UF : 0, UT : 1,

J1= -140.000 mm

};

P[45]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1558.239 mm, Y = -311.336 mm, Z = -372.101 mm,

W = -16.046 deg, P = -6.755 deg, R = -22.037 deg

GP2:

UF : 0, UT : 1,

J1= -140.000 mm

};

P[46]{

GP1:

UF : 0, UT : 1,

J1= 9.633 deg, J2= -19.594 deg, J3= -34.707 deg,

J4= 21.812 deg, J5= -61.364 deg, J6= -44.332 deg

GP2:

UF : 0, UT : 1,

J1= -120.000 mm

};

P[47]{

```
  GP1:
        UF : 0, UT : 1,
        J1=    9.168 deg, J2=   -16.084 deg,          J3=   -32.944 deg,
        J4=    10.658 deg,          J5=   -61.531 deg,          J6=   -48.701 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -120.000  mm
};
P[48]{
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1561.048  mm,       Y =  -244.305  mm,       Z =  -394.902  mm,
        W =    -8.515 deg,          P =    -3.474 deg, R =   -34.984 deg
  GP2:
        UF : 0, UT : 1,
        J1=   -68.767  mm
};
P[49]{
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1663.889  mm,       Y =  -200.188  mm,       Z =  -397.922  mm,
        W =    -2.969 deg,          P =     1.254 deg, R =   -34.020 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -105.882  mm
};
P[50]{
  GP1:
```

```
        UF : 0, UT : 1,
        J1=   10.837 deg,          J2=   -27.545 deg,          J3=   -28.113 deg,
        J4=    6.908 deg, J5=   -67.839 deg,          J6=   -45.130 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -150.000  mm
};
P[51]{
  GP1:
        UF : 0, UT : 1,
        J1=   18.096 deg,          J2=   -12.577 deg,          J3=   -41.759 deg,
        J4=   26.658 deg,          J5=   -70.273 deg,          J6=   -64.181 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -150.000  mm
};
P[52]{
  GP1:
        UF : 0, UT : 1,
        J1=   20.311 deg,          J2=    -8.446 deg, J3=   -45.330 deg,
        J4=   33.159 deg,          J5=   -70.682 deg,          J6=   -70.517 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -150.000  mm
};
P[53]{
  GP1:
        UF : 0, UT : 1,                CONFIG : 'N U T, 0, 0, 0',
```

X = 1642.225 mm, Y = -478.763 mm, Z = -257.129 mm,

W = -52.869 deg, P = .272 deg, R = -.634 deg

GP2:

UF : 0, UT : 1,

J1= -44.456 mm

};

P[54]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1694.285 mm, Y = -366.605 mm, Z = -345.360 mm,

W = -26.667 deg, P = -16.360 deg, R = -21.538 deg

GP2:

UF : 0, UT : 1,

J1= -15.304 mm

};

P[55]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1660.927 mm, Y = -200.384 mm, Z = -447.238 mm,

W = -2.969 deg, P = 1.254 deg, R = -34.020 deg

GP2:

UF : 0, UT : 1,

J1= -195.800 mm

};

P[56]{

GP1:

UF : 0, UT : 1,

J1= 20.093 deg, J2= -8.045 deg,J3= -43.209 deg,

J4= 32.881 deg, J5= -72.372 deg, J6= -69.162 deg

GP2:

UF : 0, UT : 1,

J1= -80.000 mm

};

P[57]{

GP1:

UF : 0, UT : 1,

J1= 21.307 deg, J2= -9.981 deg,J3= -47.215 deg,

J4= 44.062 deg, J5= -73.400 deg, J6= -63.285 deg

GP2:

UF : 0, UT : 1,

J1= -80.000 mm

};

P[58]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1542.063 mm, Y = -419.958 mm, Z = -311.914 mm,

W = -36.768 deg, P = -9.696 deg, R = -12.783 deg

GP2:

UF : 0, UT : 1,

J1= -21.631 mm

};

P[59]{

GP1:

UF : 0, UT : 1,

J1= 17.045 deg, J2= -5.425 deg,J3= -46.137 deg,

J4= 43.812 deg, J5= -79.584 deg, J6= -43.348 deg

GP2:

UF : 0, UT : 1,

J1= -177.077 mm

};

P[60]{

GP1:

UF : 0, UT : 1,

J1= 15.604 deg, J2= -6.695 deg,J3= -47.907 deg,

J4= 52.311 deg, J5= -70.532 deg, J6= -45.915 deg

GP2:

UF : 0, UT : 1,

J1= -170.000 mm

};

P[61]{

GP1:

UF : 0, UT : 1,

J1= 16.922 deg, J2= -8.005 deg,J3= -43.236 deg,

J4= 52.066 deg, J5= -79.254 deg, J6= -44.320 deg

GP2:

UF : 0, UT : 1,

J1= -106.100 mm

};

P[62]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1639.774 mm, Y = -499.356 mm, Z = -229.545 mm,

W = -56.978 deg, P = .445 deg, R = -.981 deg

GP2:

UF : 0, UT : 1,

J1= -46.232 mm

};

P[63]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1649.314 mm, Y = -458.448 mm, Z = -280.253 mm,

W = -51.478 deg, P = .206 deg, R = -.520 deg

GP2:

UF : 0, UT : 1,

J1= -44.923 mm

};

P[64]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1652.290 mm, Y = -422.271 mm, Z = -314.659 mm,

W = -42.926 deg, P = .560 deg, R = .501 deg

GP2:

UF : 0, UT : 1,

J1= -12.255 mm

};

P[65]{

GP1:

UF : 0, UT : 1,

J1= 16.826 deg, J2= 14.602 deg, J3= -49.901 deg,

J4= 38.499 deg, J5= -82.459 deg, J6= -61.976 deg

GP2:

UF : 0, UT : 1,

J1= -200.000 mm

};

P[66]{

GP1:

UF : 0, UT : 1,

J1= 17.761 deg, J2= 16.744 deg, J3= -53.229 deg,

J4= 60.946 deg, J5= -81.641 deg, J6= -70.492 deg

GP2:

UF : 0, UT : 1,

J1= -88.894 mm

};

P[67]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1954.507 mm, Y = -531.665 mm, Z = -156.829 mm,

W = -65.386 deg, P = -14.046 deg, R = -7.584 deg

GP2:

UF : 0, UT : 1,

J1= -88.894 mm

};

P[68]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1930.385 mm, Y = -514.605 mm, Z = -191.148 mm,

W = -62.493 deg, P = -12.696 deg, R = -6.511 deg

GP2:

UF : 0, UT : 1,

J1= -81.893 mm

};

P[69]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1916.495 mm, Y = -491.332 mm, Z = -232.487 mm,

W = -60.777 deg, P = -9.765 deg, R = -5.399 deg

GP2:

UF : 0, UT : 1,

J1= -44.040 mm

};

P[70]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1905.225 mm, Y = -464.537 mm, Z = -269.023 mm,

W = -53.269 deg, P = -9.320 deg, R = -6.982 deg

GP2:

UF : 0, UT : 1,

J1= -67.067 mm

};

P[71]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1894.750 mm, Y = -440.873 mm, Z = -294.041 mm,

W = -47.982 deg, P = -4.803 deg, R = -4.656 deg

GP2:

UF : 0, UT : 1,

J1= -124.858 mm

};

P[72]{

GP1:

UF : 0, UT : 1,

J1= 15.845 deg, J2= 7.244 deg, J3= -47.746 deg,

J4= 52.313 deg, J5= -72.555 deg, J6= -54.006 deg

GP2:

UF : 0, UT : 1,

J1= -183.648 mm

};

P[73]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1534.844 mm, Y = -450.471 mm, Z = -306.566 mm,

W = -38.981 deg, P = -10.489 deg, R = -12.818 deg

GP2:

UF : 0, UT : 1,

J1= -28.583 mm

};

P[74]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1534.935 mm, Y = -444.101 mm, Z = -296.108 mm,

W = -38.981 deg, P = -10.489 deg, R = -12.818 deg

GP2:

UF : 0, UT : 1,

J1= -20.663 mm

};

P[75]{

```
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1652.996  mm,       Y =  -440.603  mm,        Z =  -299.028  mm,
        W =   -47.584 deg,       P =     .403 deg,  R =    -.074 deg
  GP2:
        UF : 0, UT : 1,
        J1=   -24.050  mm
};
P[76]{
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1649.962  mm,       Y =  -536.749  mm,        Z =  -209.752  mm,
        W =   -59.845 deg,       P =    1.204 deg, R =   -2.063 deg
  GP2:
        UF : 0, UT : 1,
        J1=   -84.750  mm
};
P[77]{
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1665.530  mm,       Y =  -519.689  mm,        Z =  -199.834  mm,
        W =   -59.863 deg,       P =    1.635 deg, R =   -2.805 deg
  GP2:
        UF : 0, UT : 1,
        J1=   -17.842  mm
};
P[78]{
  GP1:
```

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1863.045 mm, Y = -256.483 mm, Z = -386.075 mm,

W = -11.465 deg, P = -3.954 deg, R = -28.740 deg

GP2:

UF : 0, UT : 1,

J1= -51.920 mm

};

P[79]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1865.773 mm, Y = -214.142 mm, Z = -409.716 mm,

W = -8.497 deg, P = -3.186 deg, R = -29.558 deg

GP2:

UF : 0, UT : 1,

J1= -38.847 mm

};

P[80]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1858.987 mm, Y = -209.243 mm, Z = -394.021 mm,

W = -9.081 deg, P = -1.692 deg, R = -28.919 deg

GP2:

UF : 0, UT : 1,

J1= -44.000 mm

};

P[81]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

```
        X =  1637.288  mm,       Y =  -519.868  mm,       Z =  -201.250  mm,
        W =   -59.851 deg,       P =     1.204 deg, R =    -2.063 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -106.100  mm
};
P[82]{
  GP1:
        UF : 0, UT : 1,          CONFIG : 'N U T, 0, 0, 0',
        X =  1513.774  mm,       Y =  -483.271  mm,       Z =  -234.468  mm,
        W =   157.000 deg,       P =    77.279 deg,       R =    25.101 deg
  GP2:
        UF : 0, UT : 1,
        J1=   -18.703  mm
};
P[83]{
  GP1:
        UF : 0, UT : 1,          CONFIG : 'N U T, 0, 0, 0',
        X =  1669.920  mm,       Y =  -485.752  mm,       Z =  -238.258  mm,
        W =   -52.869 deg,       P =     .272 deg,  R =     -.635 deg
  GP2:
        UF : 0, UT : 1,
        J1=   -22.924  mm
};
P[84]{
  GP1:
        UF : 0, UT : 1,          CONFIG : 'N U T, 0, 0, 0',
        X =  1657.551  mm,       Y =   -54.639  mm,       Z =  -396.752  mm,
```

W = -4.680 deg, P = .410 deg, R = -20.881 deg

GP2:

UF : 0, UT : 1,

J1= -44.000 mm

};

P[85]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1559.587 mm, Y = -196.896 mm, Z = -398.926 mm,

W = -7.655 deg, P = -5.013 deg, R = -30.028 deg

GP2:

UF : 0, UT : 1,

J1= -11.577 mm

};

P[86]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1658.271 mm, Y = -163.365 mm, Z = -397.828 mm,

W = -4.680 deg, P = .410 deg, R = -20.881 deg

GP2:

UF : 0, UT : 1,

J1= -80.914 mm

};

P[87]{

GP1:

UF : 0, UT : 1,

J1= 9.006 deg, J2= -1.094 deg, J3= -33.011 deg,

J4= 12.089 deg, J5= -62.554 deg, J6= -44.283 deg

GP2:

UF : 0, UT : 1,

J1= -111.065 mm

};

P[92]{

GP1:

UF : 0, UT : 1,

J1= 6.457 deg, J2= -12.548 deg, J3= -30.030 deg,

J4= 8.430 deg, J5= -63.566 deg, J6= -31.603 deg

GP2:

UF : 0, UT : 1,

J1= -77.449 mm

};

P[93]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1875.088 mm, Y = -365.439 mm, Z = -346.207 mm,

W = -19.561 deg, P = -13.510 deg, R = -32.554 deg

GP2:

UF : 0, UT : 1,

J1= -16.794 mm

};

P[98]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1268.891 mm, Y = -55.280 mm, Z = -460.962 mm,

W = 1.396 deg, P = -5.449 deg, R = -41.092 deg

GP2:

UF : 0, UT : 1,

J1= -240.000 mm

};

P[99]{

GP1:

UF : 0, UT : 1,

J1= -80.513 deg, J2= -42.854 deg, J3= -5.500 deg,

J4= -.092 deg, J5= -74.501 deg, J6= 90.529 deg

GP2:

UF : 0, UT : 1,

J1= -3.000 mm

};

P[100]{

GP1:

UF : 0, UT : 1,

J1= -108.003 deg, J2= 28.024 deg, J3= -9.389 deg,

J4= .952 deg, J5= -78.990 deg, J6= 110.036 deg

GP2:

UF : 0, UT : 1,

J1= -3.000 mm

};

P[101]{

GP1:

UF : 0, UT : 1,

J1= -107.017 deg, J2= 33.914 deg, J3= -15.628 deg,

J4= 1.007 deg, J5= -72.769 deg, J6= 108.934 deg

GP2:

UF : 0, UT : 1,

J1= -3.000 mm

};

P[104]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1663.631 mm, Y = -250.977 mm, Z = -462.852 mm,

W = -13.484 deg, P = -1.547 deg, R = -22.128 deg

GP2:

UF : 0, UT : 1,

J1= -95.278 mm

};

P[105]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1659.408 mm, Y = -260.836 mm, Z = -444.423 mm,

W = -10.998 deg, P = -1.797 deg, R = -24.907 deg

GP2:

UF : 0, UT : 1,

J1= -91.450 mm

};

P[106]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1659.408 mm, Y = -260.836 mm, Z = -444.423 mm,

W = -10.997 deg, P = -1.797 deg, R = -24.176 deg

GP2:

UF : 0, UT : 1,

J1= -116.200 mm

};

P[107]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1659.958 mm, Y = -329.801 mm, Z = -427.707 mm,

W = -17.238 deg, P = -2.631 deg, R = -17.801 deg

GP2:

UF : 0, UT : 1,

J1= -89.291 mm

};

P[108]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1554.848 mm, Y = -324.703 mm, Z = -419.715 mm,

W = -16.046 deg, P = -6.755 deg, R = -22.037 deg

GP2:

UF : 0, UT : 1,

J1= -140.000 mm

};

P[109]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1556.164 mm, Y = -256.986 mm, Z = -447.700 mm,

W = -8.515 deg, P = -3.474 deg, R = -31.588 deg

GP2:

UF : 0, UT : 1,

J1= -120.000 mm

};

P[110]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1553.673 mm, Y = -255.285 mm, Z = -447.881 mm,

W = -8.515 deg, P = -3.474 deg, R = -34.514 deg

GP2:

UF : 0, UT : 1,

J1= -120.000 mm

};

P[111]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1645.241 mm, Y = -478.136 mm, Z = -360.394 mm,

W = -47.159 deg, P = .379 deg, R = -11.272 deg

GP2:

UF : 0, UT : 1,

J1= -117.640 mm

};

P[112]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1814.109 mm, Y = -560.875 mm, Z = -291.603 mm,

W = -58.649 deg, P = -9.013 deg, R = -15.234 deg

GP2:

UF : 0, UT : 1,

J1= -200.000 mm

};

P[113]{

```
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1891.182  mm,      Y =  -543.901  mm,      Z =  -268.464  mm,
        W =   -58.890 deg,      P =   -10.426 deg,      R =   -13.296 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -207.342  mm
};
P[114]{
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1924.719  mm,      Y =  -537.079  mm,      Z =  -221.021  mm,
        W =   -62.875 deg,      P =   -12.847 deg,      R =    -9.735 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -127.499  mm
};
P[115]{
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1924.815  mm,      Y =  -530.794  mm,      Z =  -201.339  mm,
        W =   -62.434 deg,      P =   -14.599 deg,      R =    -7.513 deg
  GP2:
        UF : 0, UT : 1,
        J1=   -44.605  mm
};
P[116]{
  GP1:
```

```
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1916.225  mm,       Y =  -515.000  mm,        Z =  -229.541  mm,
        W =   -60.487 deg,       P =    -8.764 deg, R =    -7.196 deg
  GP2:
        UF : 0, UT : 1,
        J1=   -44.039  mm
};
P[117]{
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1907.028  mm,       Y =  -493.112  mm,        Z =  -265.368  mm,
        W =   -59.126 deg,       P =    -4.741 deg, R =    -7.032 deg
  GP2:
        UF : 0, UT : 1,
        J1=   -95.169  mm
};
P[118]{
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
        X =  1895.279  mm,       Y =  -458.629  mm,        Z =  -308.794  mm,
        W =   -54.960 deg,       P =    -4.286 deg, R =    -4.230 deg
  GP2:
        UF : 0, UT : 1,
        J1=   -49.123  mm
};
P[119]{
  GP1:
        UF : 0, UT : 1,              CONFIG : 'N U T, 0, 0, 0',
```

```
        X =  1891.590  mm,      Y =  -467.050  mm,      Z =  -316.897  mm,
        W =   -49.434 deg,      P =    -5.649 deg, R =    -5.424 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -148.259  mm
};
P[122]{
  GP1:
        UF : 0, UT : 1,         CONFIG : 'N U T, 0, 0, 0',
        X =  1889.833  mm,      Y =  -467.859  mm,      Z =  -327.402  mm,
        W =   -52.031 deg,      P =    -3.532 deg, R =    -3.855 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -183.648  mm
};
P[123]{
  GP1:
        UF : 0, UT : 1,         CONFIG : 'N U T, 0, 0, 0',
        X =  1716.986  mm,      Y =  -447.359  mm,      Z =  -349.475  mm,
        W =   -53.544 deg,      P =     9.102 deg, R =     1.486 deg
  GP2:
        UF : 0, UT : 1,
        J1=  -183.648  mm
};
P[124]{
  GP1:
        UF : 0, UT : 1,         CONFIG : 'N U T, 0, 0, 0',
        X =  1673.792  mm,      Y =  -437.900  mm,      Z =  -383.819  mm,
```

W = -53.544 deg, P = 9.102 deg, R = 1.485 deg

GP2:

UF : 0, UT : 1,

J1= -189.258 mm

};

P[125]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1559.292 mm, Y = -456.539 mm, Z = -373.948 mm,

W = -53.083 deg, P = 15.407 deg, R = 3.025 deg

GP2:

UF : 0, UT : 1,

J1= -183.648 mm

};

P[126]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1559.989 mm, Y = -202.407 mm, Z = -456.518 mm,

W = -8.515 deg, P = -3.474 deg, R = -20.720 deg

GP2:

UF : 0, UT : 1,

J1= -120.000 mm

};

P[127]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1485.762 mm, Y = -426.598 mm, Z = -311.268 mm,

W = -48.719 deg, P = 28.633 deg, R = 14.555 deg

GP2:

UF : 0, UT : 1,

J1= -228.198 mm

};

P[128]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1409.364 mm, Y = -418.344 mm, Z = -249.216 mm,

W = -49.977 deg, P = 28.951 deg, R = 22.575 deg

GP2:

UF : 0, UT : 1,

J1= -228.198 mm

};

P[129]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1310.967 mm, Y = -395.576 mm, Z = -222.747 mm,

W = -48.767 deg, P = 37.134 deg, R = 24.305 deg

GP2:

UF : 0, UT : 1,

J1= -248.350 mm

};

P[131]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1237.919 mm, Y = -451.094 mm, Z = -136.606 mm,

W = -36.017 deg, P = 31.027 deg, R = 44.225 deg

G UF : 0, UT : 1,

J1= -248.350 mm

};

P[132]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1709.754 mm, Y = -373.019 mm, Z = -353.315 mm,

W = -26.027 deg, P = -13.659 deg, R = -20.481 deg

GP2:

UF : 0, UT : 1,

J1= -32.389 mm

};

P[133]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1763.518 mm, Y = -381.572 mm, Z = -354.144 mm,

W = -24.600 deg, P = -8.725 deg, R = -18.277 deg

GP2:

UF : 0, UT : 1,

J1= -59.888 mm

};

P[146]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1526.248 mm, Y = -402.945 mm, Z = -232.134 mm,

W = -156.526 deg, P = 79.329 deg, R = 72.610 deg

GP2:

UF : 0, UT : 1,

J1= -243.847 mm

};

P[147]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1567.145 mm, Y = -439.948 mm, Z = -227.394 mm,

W = 152.880 deg, P = 79.001 deg, R = 21.212 deg

GP2:

UF : 0, UT : 1,

J1= -243.846 mm

};

P[151]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1882.827 mm, Y = -385.517 mm, Z = -334.853 mm,

W = -24.422 deg, P = -10.582 deg, R = -29.676 deg

GP2:

UF : 0, UT : 1,

J1= -45.000 mm

};

P[152]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1638.508 mm, Y = -518.442 mm, Z = -199.033 mm,

W = -59.851 deg, P = 1.204 deg, R = -2.063 deg

GP2:

UF : 0, UT : 1,

J1= -106.100 mm};P[153]{ GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1993.981 mm, Y = -532.155 mm, Z = -156.838 mm,

W = -69.576 deg, P = -14.046 deg, R = -7.584 deg

GP2:

UF : 0, UT : 1,

J1= -12.252 mm

};

P[156]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1561.848 mm, Y = -156.909 mm, Z = -397.421 mm,

W = -5.807 deg, P = -2.178 deg, R = -20.847 deg

GP2:

UF : 0, UT : 1,

J1= -44.000 mm

};

P[158]{

GP1:

UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',

X = 1871.376 mm, Y = -366.312 mm, Z = -357.069 mm,

W = -24.862 deg, P = -8.471 deg, R = -24.901 deg

GP2:

UF : 0, UT : 1,

J1= -55.970 mm

};P[161]{ GP1: UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0', X = 1518.261 mm, Y = -477.056 mm,
Z = -268.951 mm, W = 157.000 deg, P = 77.279 deg, R = 25.101 deg

GP2: UF : 0, UT : 1, J1= -7.896 mm};P[162]{ GP1: UF : 0, UT : 1, CONFIG : 'N U T, 0, 0, 0',
X = 1574.649 mm, Y = -427.254 mm, Z = -230.0 W = 156.999 deg, P = 77.280
deg, R = 25.099 deg GP2:

UF : 0, UT : 1, J1= -243.846 mm} /END